Research Roadmap

Multi-Agent Reinforcement Learning for Adaptive Traffic Signal Optimization in Smart Cities

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Midhun Chandrasekhar
B Tech, MBA
Dubai
United Arab Emirates

1. Overview

This research follows a **phased, iterative approach** to design, validate, and scale a Multi-Agent Reinforcement Learning (MARL) framework for traffic signal optimization.

The roadmap emphasizes:

- Controlled experimentation
- Incremental complexity
- Measurable validation at each stage

2. Phase 1 — Simulation & Baseline (Completed)

Objective

Establish a reliable experimental foundation for traffic modeling and evaluation.

Key Activities

- Design and implement grid-based traffic simulation environment
- Model vehicle flow and intersection behavior
- Implement fixed-time signal control system
- Develop evaluation metrics and visualization dashboard

Outcome

- Functional simulation platform
- Baseline benchmark for performance comparison
- Initial validation of traffic modeling assumptions

3. Phase 2 — MARL System Design (Ongoing)

Objective

Define and implement the core Multi-Agent Reinforcement Learning architecture.

Key Activities

- Define agent structure for each intersection
- Design state, action, and reward representations
- Select initial RL algorithms (e.g., Q-Learning / DQN)
- Implement decentralized learning framework

Expected Outcome

- Working MARL prototype
- Initial training results
- Early comparison with baseline system

4. Phase 3 - Training & Optimization

Objective

Improve learning performance, stability, and efficiency of agents.

Key Activities

- Hyperparameter tuning
- Reward function refinement
- Exploration vs exploitation balancing
- Convergence analysis

Expected Outcome

- Stable and efficient learning agents
- Measurable improvement over baseline system
- Consistent performance across multiple runs

5. Phase 4 - Controlled Scalability Evaluation

Objective

Evaluate system performance under increasing network complexity.

Key Activities

- Extend simulation from 4×4 to 6×6 / 8×8 grids
- Analyze coordination between agents
- Measure performance degradation or stability
- Identify scalability limitations

Expected Outcome

- Validation of decentralized approach under larger systems
- Insights into coordination dynamics
- Evidence of scalability potential

6. Phase 5 - Advanced Scenarios & Realism

Objective

Introduce more realistic and dynamic traffic conditions.

Key Activities

- Simulate non-uniform traffic patterns
- Introduce dynamic traffic surges
- Model heterogeneous traffic flows
- Explore irregular road network structures

Expected Outcome

- Improved robustness of MARL system
- Better alignment with real-world traffic behavior
- Enhanced generalization capability

7. Phase 6 - Real-World Integration (Exploratory)

Objective

Bridge the gap between simulation and real-world applicability.

Key Activities

- Integrate real-world traffic datasets (if available)
- Align simulation parameters with real traffic conditions
- Evaluate transferability of learned policies

Expected Outcome

- Preliminary validation in real-world contexts
- Identification of deployment challenges
- Foundation for applied research or pilot studies

8. Phase 7 - Research Contribution & Publication

Objective

Formalize findings into academic contributions.

Key Activities

- Document experimental results
- Compare with existing methods in literature
- Prepare research papers for conferences/journals
- Open-source simulation and framework

Expected Outcome

- Publishable research output
- Reproducible experimental framework
- Contribution to intelligent transportation research

9. LongTerm Vision

This research aims to evolve into:

- A scalable framework for intelligent traffic control
- Integration with IoT-enabled smart city infrastructure
- Support for autonomous and connected vehicle ecosystems

10. Summary

The roadmap follows a **progressive validation strategy**, starting from controlled simulation and moving toward scalable and realistic scenarios.

Each phase builds upon the previous one, ensuring:

- Strong experimental foundation
- Measurable research contributions
- Practical relevance to real-world traffic systems